

Diego Atencia

Machine Learning Engineer — Banking Data & GenAI

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SUMMARY

Mathematician (UAM) turned ML engineer with 3+ years across banking data and AI: regulatory data pipelines (Oracle, T-SQL), IRB credit-risk modeling (LGD), and production GenAI (LoRA fine-tuning, RAG, AWS). Open-source contributor to LLM training infrastructure (Liger-Kernel, TorchRL).

EXPERIENCE

Senior Data Scientist

Jun 2025 – Present

PwC — Madrid

- Data engineering and analysis on the LGD database of a top-4 Spanish bank, where small errors have significant impact on the bank's reserved capital and earnings.
- Developed Power Automate/Copilot Studio workflows integrating LGD model outputs into structured Word reporting pipelines for audit work.
- Developing a data-quality-control (DQC) automation tool, currently at proof-of-concept stage.
- Contributed to landing 8 contract renewals with this client; integrated 2 junior engineers now working full-time on the account.

Data Scientist

Nov 2023 – Jun 2025

PwC — Madrid

- Worked on stress testing for a PD (Probability of Default) climate-risk application in Excel.

SQL & PL/SQL Developer

Oct 2022 – Nov 2023

Álamo Consulting — Madrid

- Owned regulatory reporting pipelines (FINREP, COREP, SIRBE/CIRBE) for an external banking entity using Oracle and T-SQL.
- Built scalable transformations within a large regulatory codebase, ensuring accurate and timely data delivery.

OPEN-SOURCE CONTRIBUTIONS

Liger-Kernel — github.com/linkedin/Liger-Kernel

PR #1128 (under review)

Production CUDA kernel library for LLM training, used at scale across industry

- Debugging a distributed-training bug in **LigerTiledSwiGLUMLP**: FSDP's post-backward reshard hooks were triggered prematurely inside a tiling loop, corrupting parameter shards across GPUs.
- Adapting axolotl's gradient-accumulator approach as the fix.

TorchRL — github.com/pytorch/rl

Discussion #3538

PyTorch RL library (Meta)

- Proposing and implementing MC-PILCO: a model-based RL algorithm using Gaussian Process world models and Monte Carlo rollouts.

LEANN

Issue #41

- Proposed a hybrid BM25+RRF retrieval pipeline over the neural-only architecture to improve recall on sparse queries.

SELECTED PROJECTS

Banking-Regulation LLM — fine-tuned model for Spanish banking regulation Q&A

2025–2026

- Fine-tuned Qwen2.5-7B with LoRA on regulatory corpora; model published on Hugging Face.
- Built the full application around it: RAG pipeline and AWS deployment (ECS Fargate, RDS/pgvector, ALB); served real user queries during early access.

Hand Prosthesis Control with Deep RL — blog

M.Sc. thesis

- Multimodal RL agent (PPO/DDPG) integrating stereoscopic vision and sensor data for real-time prosthetic hand control in MuJoCo simulation.

EDUCATION

M.Sc. in Artificial Intelligence (8.67/10)
International University of Valencia
Thesis: *Optimizing low-cost hand prosthesis control using Deep Reinforcement Learning*, multimodal architecture integrating stereoscopic vision and tactile sensors.

Sep 2024 – Oct 2025

B.Sc. in Mathematics
Autonomous University of Madrid (UAM)
Final thesis: Weak solutions to Navier–Stokes equations (9/10). Founder of university AI student association (70+ members). Rugby team member.

Sep 2018 – Jun 2023

TECHNICAL SKILLS

Languages	Python (expert), C/C++ (strong), SQL (expert)
Systems & Cloud	AWS, Docker, Linux, Git/GitHub, CI/CD principles, distributed training (FSDP/DDP), CUDA kernels
Databases	Oracle, PostgreSQL, ChromaDB (vector), SAS
ML / GenAI	PyTorch, TensorFlow, Scikit-learn, HuggingFace, LoRA/QLoRA, RAG pipelines
CS Fundamentals	Data structures & algorithms, OOP, distributed systems, debugging complex systems, code review

LANGUAGES

Spanish (native) English (fluent, professional working proficiency) Mandarin Chinese (learning)